

SOLUTION & APPROACH · PROBLEM DEFINITION

HOURS OF DIGGING FOR A 15-MINUTE MEETING.

The data already exists. It lives in three places, and the rep does the synthesis by hand, the night before.

MARKET INTELLIGENCE

Who to call. Not what to say.

PRODUCT KNOWLEDGE

What Tempus can claim. Not to whom.

CRM NOTES

The concern raised last time. Not the answer to it.

THE ASPECT I AM SOLVING

The **synthesis**, not the plumbing. Collection and CRM already work. What does not exist is the last mile: turning three static sources into one thing a rep can say out loud, and defend when the doctor pushes back.

WHAT I BUILT, AND WHAT I DID NOT

THE HONEST VERSION OF THE SCOPE.

ASSUMPTIONS

- 01

"Provider" means the oncologist

The case study's word, and standard healthcare usage. The meeting is with a doctor or a CMO.
- 02

Impact is a magnitude, not a guess

"Ordered by impact" asks how much is at stake, and the data carries exactly one magnitude: patient population. That is the sort. What it does *not* carry is propensity, so I refuse to invent one.
- 03

The prototype begins after ingestion

Committed snapshots. No Salesforce sync, no live scraping, no database.
- 04

Product facts are public-page only

So no pricing or coverage fact exists, and a cost question routes to a human. An eval check asserts that gap stays a gap.

DELIBERATELY NOT BUILT

- x

A chat interface

The rep has fifteen minutes and already knows what they need. Making them prompt their way to it hands the work back to them and returns something different every time. **One click, one brief, read in thirty seconds.**
- x

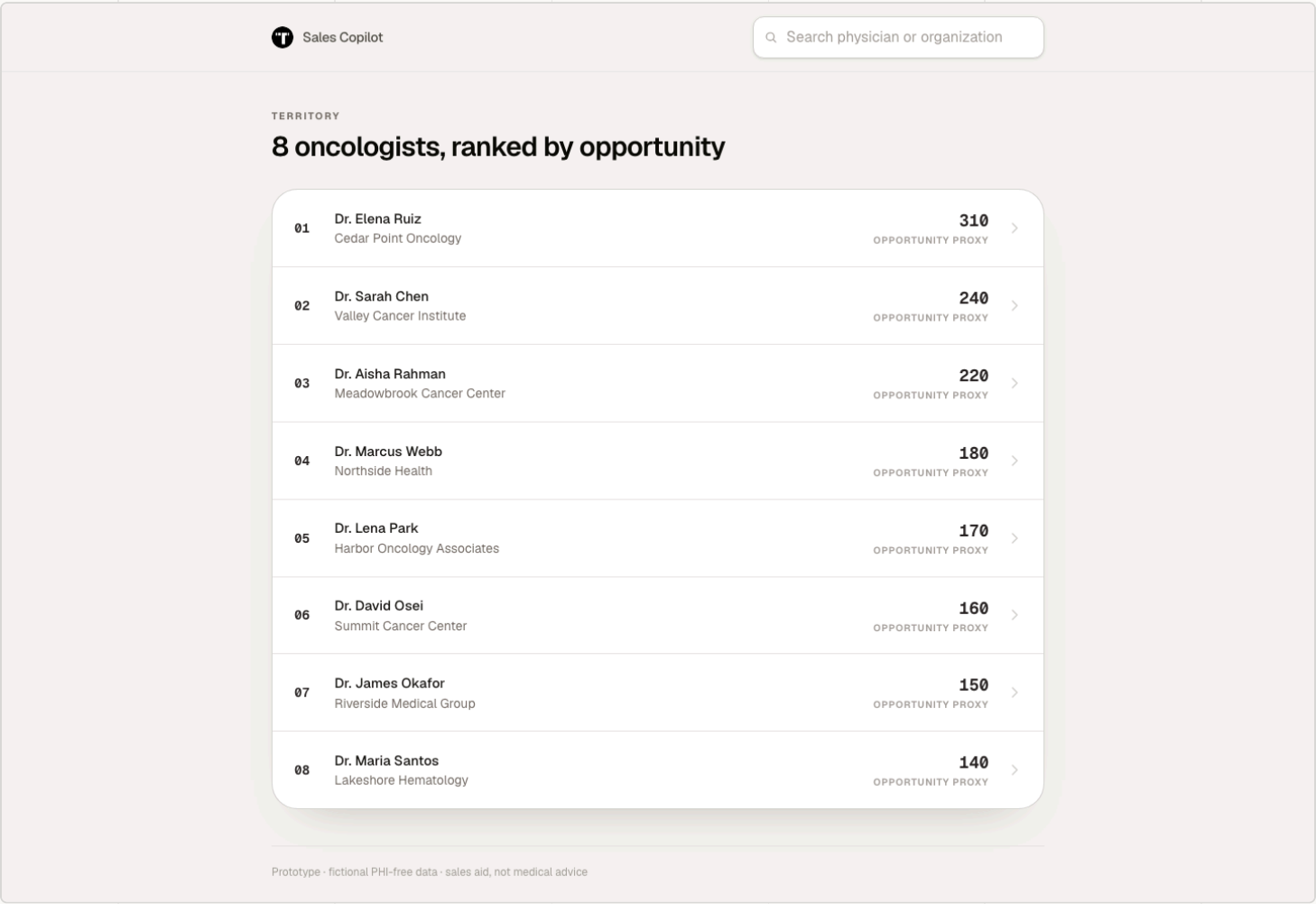
"Why now"

The three supplied sources are timeless. A defensible answer needs a dated catalyst feed. I built one and cut it rather than manufacture urgency.

THE SOLUTION

ONE PLACE.
EVERYTHING THE REP
NEEDS.

Open a doctor and the brief is generated on the spot.
Seconds, not an evening.



RANKED BY LIKELY PATIENT POPULATION

THE TRADEOFF: ONE HONEST SORT, NOT A SALES SCORE

Expected value = impact × propensity. "Ordered by impact" asks for the first term. The data supplies exactly one magnitude, *likely patient population*, so that is the sort, and the column says *directional proxy* out loud.

Propensity is the term nobody was given. No conversion history, no won/lost, no engagement. A 1-100 "lead score" fits weights to eight unlabeled rows across eight one-off concerns. It would look calibrated and be unfalsifiable.

The CRM signal is used where it can be checked. It picks the products, the facts, the objection, the script – all validated against the vault. It never becomes a number, because in a ranking nothing could check it.

PER DOCTOR

THE OBJECTION,
ANSWERED. THE PITCH,
TIMED.

- 01 Answers the CRM concern with a real metric
- 02 65–80 words. Thirty seconds, enforced
- 03 Every claim links to its source

- ✓ xF results are typically expected within 7 days of specimen retrieval.
Turnaround Time ↗
- ✓ Tempus describes an automatic conversion option from xT CDx to xF or xF+ when tumor tissue is insufficient.
Ordering Flexibility ↗

OBJECTION HANDLER

The xF liquid-biopsy panel results are typically expected within 7 days of specimen retrieval. This is a typical expectation and not a guaranteed turnaround or a service-level agreement. For cases where tissue is insufficient, we also offer an automatic conversion option from xT CDx to xF to help streamline the testing process for your patients.

30-SECOND MEETING SCRIPT

74 words

Dr. Chen, thank you for sharing your focus on time-sensitive first-line decisions for your NSCLC patients. Regarding your question on liquid-biopsy turnaround, xF results are typically expected within 7 days of specimen retrieval. This typical expectation helps provide timely molecular insights when tissue is limited. We also offer an automatic conversion option from xT CDx to xF to support your workflow. Could we discuss how this typical turnaround fits into your current clinical process?

TECHNICAL APPROACH

THREE SOURCES IN. ONE GROUNDED BRIEF OUT.

01 LOAD This doctor's CRM note. The product fact vault.	02 BUILD CONTEXT One bounded package. No other doctor's data can enter it.	03 GENERATE One structured call. The schema is enforced at the API.	04 VALIDATE Every claim checked against the vault. Blocking.	05 SHIP OR BLOCK One repair attempt. Then it ships, or the rep is told.
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THE TRADEOFF: NO RAG, NO AGENT LOOP

The whole vault is ~6,500 tokens. It fits in one prompt with room to spare, so there is nothing to retrieve *from*. A retriever here can only lose, by dropping the one fact that answered the question.

The model never picks a tool or decides to **retry**. Code does. Every run is reproducible, and every failure is a test rather than a mystery.

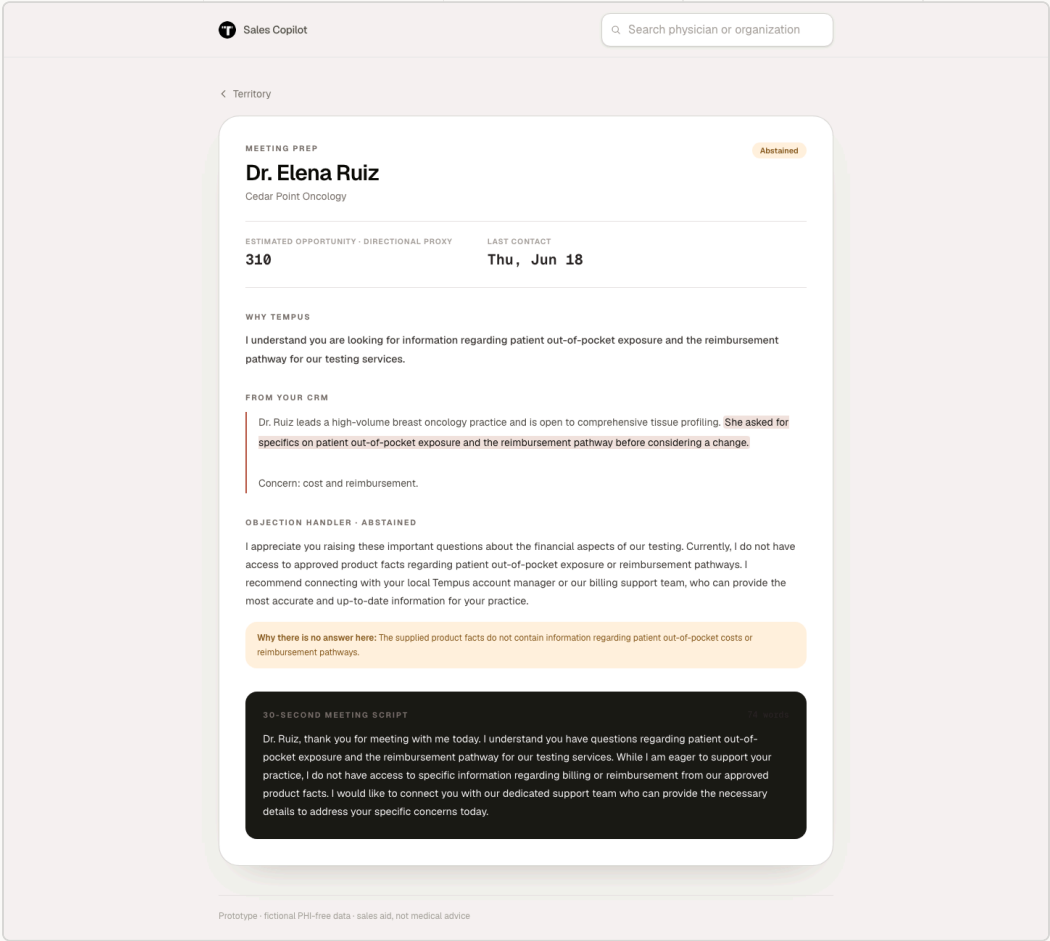
The seam is already drawn. Retrieval drops in behind the knowledge interface the day the corpus earns it. Nothing above it changes.

GROUNDING

EVERY CLAIM IS CHECKED BEFORE THE REP SEES IT.

- ✓ The CRM quote is verbatim
- ✓ Every number traces to a cited fact
- ✓ Qualifiers survive
 - "Typically within 7 days" never becomes "within 7 days."
- ✓ No guarantee, competitor, or pricing claim

If a check fails, the brief is blocked. The rep is never handed a claim we cannot back.



NO PRICING FACT EXISTS, SO IT SAYS SO AND ROUTES TO A HUMAN

HOW I ARRIVED AT IT

SPEC FIRST. DETERMINISTIC LAYERS. THEN THE MODEL.

- 01

Wrote the spec before any code

SPEC.md settled the brief contract, the validation rules, and the schemas. It is what Claude Code and Codex built against. The spec was the prompt.
- 02

Built everything that needs no model first

Loaders, the ranking, the validator. All testable without a single API call.
- 03

Added the model last, constrained by the validator it would be graded on
- 04

Iterated the prompt against an eval, not against taste

Four versions, each driven by a measured failure.

TOOLS

Claude Code

Codex

Skills

Gemini

Next.js

TypeScript

Vercel

GitHub

THE REPAIR PROMPT

```
# A rejected draft is not redrawn. It is shown
# exactly which checks it failed.

Your previous brief was REJECTED.

  x qualifiers_preserved: xf-turnaround-01
    You wrote "within 7 days".
    The fact says "typically within 7 days".
    The qualifier is not optional.

  x numbers_trace_to_cited_facts: 648
    You cited no fact stating 648.

Rewrite the brief. Fix only these.
```

Every version, and the measured failure behind it:
prompts/meeting-prep-v1..v4.md + prompts/CHANGELOG.md.

How I refined it

The Eval Told Me What To Fix.

- 01

Declare the right answer first

For each of the eight doctors I wrote down what the copilot *should* do, and which exact fact it must cite, before generating anything.
- 02

Generate, and let the validator reject

Anything that misses the declared answer fails. No human judgement in the loop.
- 03

Read the failure, not the score

Not "bad output." It says: *you dropped "typically" from xf-turnaround-01.*
- 04

Fix the prompt. Re-run. Measure.

Four versions, each driven by a named failure.

The prompt on its own	7-8 / 8
What the rep actually gets	8 / 8

I report both. The second number includes the one retry, so quoting it alone would credit the prompt for what the retry fixed.

Four times, the harness was wrong

THE MODEL WROTE

"This is a typical expectation, not a guaranteed turnaround."

THE VALIDATOR BLOCKED IT

for using the word "guaranteed."

The model was doing exactly what the rule existed to enforce, and got punished for saying the safe thing out loud. **So I fixed the check, not the model.** Negated phrases are now stripped before the prohibition runs. "We guarantee 7 days" is still blocked.

A validator you never correct is a validator you do not trust.